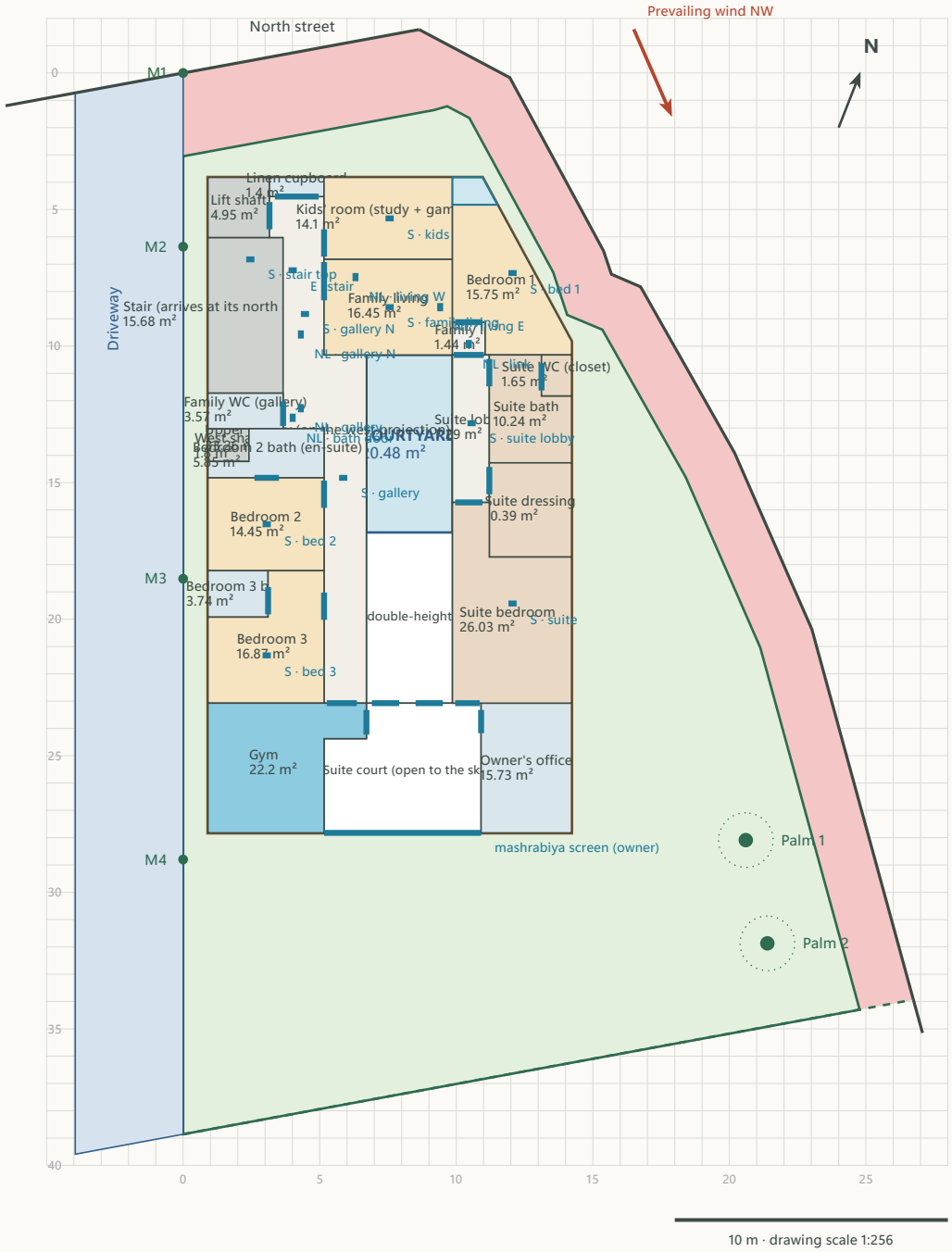



# Taif / Palm House

## First floor V8.1 · family part + the L (office, gym, suite court, master suite)

Taif / Palm House V7.1 · rooms in data/rooms_ff.csv · design and furniture in studies/ff_family.typ and studies/ff_suite.typ · schedule, light check and the split projection on the next pages

**First floor V8.1.** The gallery runs on a **1.55 m west projection** along the courtyard and the double-height void (x 4.25–5.8, y 6.5–19.25), on an **edge beam and 3 columns** at x 5.8, each in an existing wall or glass line; the first-floor west wall stands on the GF west wall (0.785 m from the M line). Courtyard sky and void 3.15 m on axes. West rooms 4.25 m and gallery 1.55 m on axes: **bedroom 2**, **bedroom 3 with its own bath**, the **family bath** by the west shaft. North band: **kids' room (study + gaming)** and the **family living** on the courtyard edge. **Bedroom 1** is one room on the straight y 6.5 line. The **south wing v2** (1 Nov 2026): the **gym** on the west band (the treadmill with its 2.0 / 0.5 m zone, an acoustic double wall to bedroom 3, an entrance nib at the gallery end), the **suite court** from the gym to the office (≈ 25 m², open to the sky, the three court doors on a 1.2 m strip along its north edge) in front of the clerestory windows, the **owner's office** on the east beside the suite (no door to the bedroom; reached from the court only; visitors are met in the men's majlis), and the **master suite** (lobby, luxury bath with WC closet, walk-in dressing, bedroom). No east duct. Stair (its arrival at the north end since the flip), lift and west shaft rise from the ground floor. Colours: suite (beige) · family (yellow) · baths, office and linen (blue-grey) · gym (blue) · circulation (light) · fixed (grey) · courtyard, void and court (white / blue). Blue marks: doors and the clerestory windows. Areas: available 245.72 m² · rooms 220.43 m² (circulation 41.99 m²) · suite court 25.3 m².
✓ **data/rooms_ff.csv matches the two studies room for room.**

✓ **All checks passed**

## Areas

| | |
|---|---|
| Envelope | **667.87 m²** |
| Building limit (envelope unless set) | **309.68 m²** |
| Other open voids | **44.98 m²** |
| Courtyard (open to sky) | **20.48 m²** |
| Rooms placed (21) | **242.85 m²** |
| Limit minus courtyard and voids | **244.22 m²** |
| Air channel + duct | **1.37 m²** |
| Not yet assigned | **1.37 m²** |

## Orientation

| | |
|---|---|
| True compass vs the M line | **21.4°** |
| NW wind vs the M line | **23.6°** |

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

# Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

**Envelope corners**

| Corner | x | y |
|---|---|---|
| V1 | 0 | 38.86 |
| V2 | 0 | 3.05 |
| V3 | 9.18 | 1.36 |
| V4 | 9.67 | 1.22 |
| V5 | 10.48 | 1.65 |
| V6 | 13.55 | 7.3 |
| V7 | 14.06 | 8.86 |
| V8 | 15.35 | 9.4 |
| V9 | 18.4 | 14.8 |
| V10 | 21.13 | 21.04 |
| V11 | 24.77 | 34.3 |

**Rooms**

| Name | Frame | w × d (m) | Area |
|---|---|---|---|
| Stair (arrives at its north end · flipped) | M | 2.75 × 5.7 | 15.68 m² |
| Lift shaft | M | 2.25 × 2.2 | 4.95 m² |
| West shaft | M | 1.5 × 1.2 | 1.8 m² |
| Upper gallery (on the west projection) | M | 3.55 × 18.55 | 33.26 m² |
| Linen cupboard | M | 2 × 0.7 | 1.4 m² |
| Family WC (gallery) | M | 2.75 × 1.3 | 3.57 m² |
| Bedroom 2 bath (en-suite) | M | 4.25 × 1.8 | 5.85 m² |
| Bedroom 2 | M | 4.25 × 3.4 | 14.45 m² |
| Bedroom 3 bath | M | 2.2 × 1.7 | 3.74 m² |
| Bedroom 3 | M | 4.25 × 4.85 | 16.87 m² |
| Kids' room (study + gaming) | M | 4.7 × 3 | 14.1 m² |
| Family living | M | 4.7 × 3.5 | 16.45 m² |
| Family link | M | 1.2 × 1.2 | 1.44 m² |
| Bedroom 1 | M | 4.361 × 5.5 | 15.75 m² |
| Owner's office | M | 3.311 × 4.75 | 15.73 m² |
| Gym | M | 5.8 × 4.75 | 22.2 m² |
| Suite lobby | M | 1.35 × 5.4 | 7.29 m² |
| Suite bath | M | 3.011 × 3.95 | 10.24 m² |
| Suite WC (closet) | M | 1.1 × 1.5 | 1.65 m² |
| Suite dressing | M | 3.011 × 3.45 | 10.39 m² |
| Suite bedroom | M | 4.361 × 7.35 | 26.03 m² |

# First floor V6.3 · room schedule

wall centre lines (axes) · clear ≈ axes minus half a wall on every edge (from the studies) · CSV coordinates

| Room | Part | Axes w × d (m) | Axes m² | ≈ clear m² |
|---|---|---|---|---|
| Upper gallery (on the west projection) | family | polygon | 33.26 | 29.95 |
| Linen cupboard | family | 2 × 0.7 | 1.4 | – |
| Family WC (gallery) | family | 2.75 × 1.3 | 3.57 | 2.73 |
| Bedroom 2 bath (en-suite) | family | polygon | 5.85 | 4.63 |
| Bedroom 2 | family | 4.25 × 3.4 | 14.45 | 12.84 |
| Bedroom 3 bath | family | 2.2 × 1.7 | 3.74 | 2.92 |
| Bedroom 3 | family | polygon | 16.87 | 14.97 |
| Kids' room (study + gaming) | family | 4.7 × 3 | 14.1 | 12.44 |
| Family living | family | 4.7 × 3.5 | 16.45 | 15.16 |
| Family link | family | 1.2 × 1.2 | 1.44 | 1.08 |
| Bedroom 1 | family | polygon | 15.75 | 13.83 |
| Owner's office | L | 3.31 × 4.75 | 15.73 | 13.91 |
| Gym | L | polygon | 22.2 | 19.87 |
| Suite court (open to the sky) | L | polygon | 25.3 | – |
| Suite lobby | L | 1.35 × 5.4 | 7.29 | 5.94 |
| Suite bath | L | polygon | 10.24 | 8.79 |
| Suite WC (closet) | L | 1.1 × 1.5 | 1.65 | 1.09 |
| Suite dressing | L | 3.01 × 3.45 | 10.39 | 9.01 |
| Suite bedroom | L | polygon | 26.03 | 23.56 |
| **Rooms (indoor)** | | | **220.43** | |
| **Suite court (open)** | | | **25.3** | |
| Stair, lift, west shaft, air tower | | | 23.8 | |

## Key numbers

- **Master suite:** 55.6 m² indoors (lobby, bath + WC closet, dressing, bedroom), + the suite court 25.3 m².
- **Circulation:** 41.99 m² (gallery, family link, suite lobby; the stair arrives on the gallery's north part since the flip).
- **Courtyard sky** 3.15 × 6.5 m and **double-height void** 3.15 × 6.25 m on axes; the ground-floor courtyard and living keep their full floor.
- **West rooms:** 4.25 m on axes (≈ 4.025 m clear); **gallery** 1.55 m (≈ 1.42 m clear); **bedroom 2's bath** 4.25 m wide (3 Nov 2026).
- **Edge beam** on x 5.8 (spans 6.5 and 6.25 m) on 3 columns; the FF west wall on the GF west wall, 0.785 m from the M line.

## Where the detail is

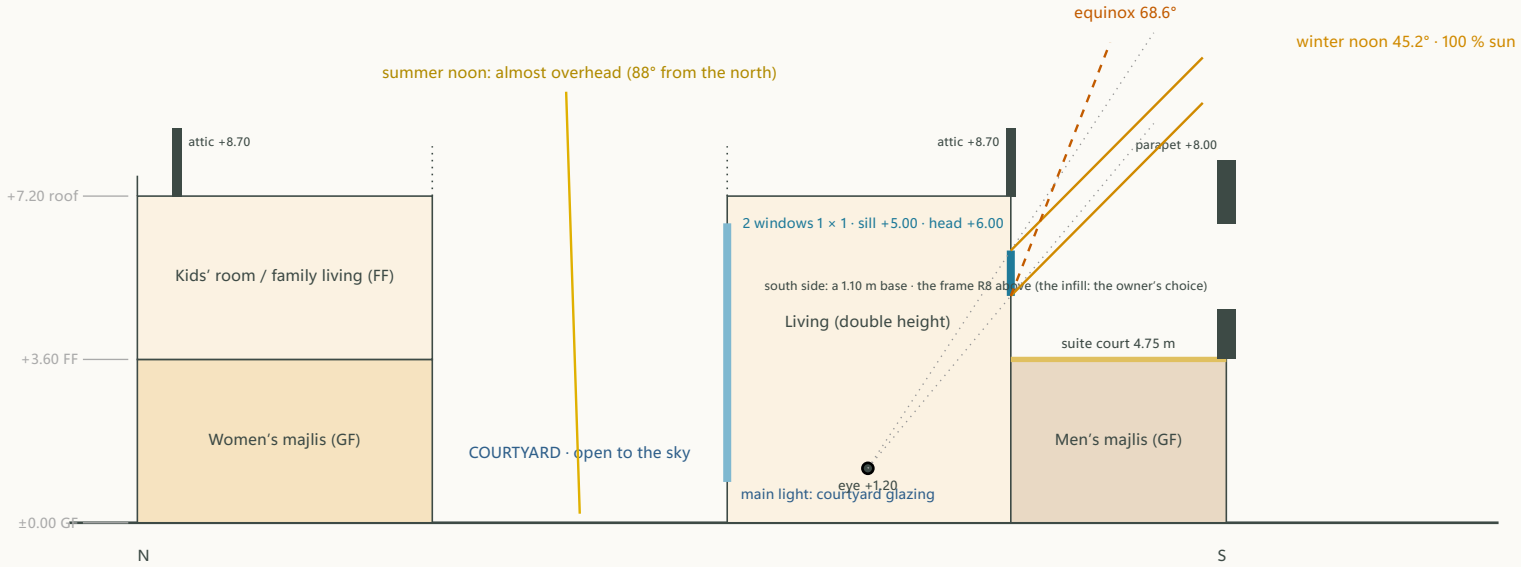
- **Furniture, doors, windows, drains, walking distances, routes ≥ 1.0 m and room checks:** studies/ff_family.typ (part 1) and studies/ff_suite.typ (part 2).
- **Decisions with reasons, construction notes, changelog:** docs/PROJECT_V6.md · **open items:** docs/OPEN_ITEMS.md.

## Open items (first floor)

- Engineer: the edge beam (spans 6.5 and 6.25 m) and the 3 columns at x 5.8: sizes and foundations.
- Engineer: loads and vibration of the gym and the office over the men's majlis; the y 3.0 partition over the women's majlis.
- MEP: the stack outside the east wall (suite bath, vent stack), the kitchen extract through the east wall, the hood south of y ≈ 7.5.
- The clerestory sill against the suite court floor (eye height, or obscured glass); a daylight check of the courtyard.

# First floor V8.1 · light check · the double-height living’s south windows

section north → south through the middle band (CSV x ≈ 7.3) · levels to confirm: GF ±0.00 · FF +3.60 · roof +7.20 · parapets 1.1




| Behind the 4.75 m suite court | Winter noon sun on the windows | Equinox noon sun | Sky seen from the eye |
|---|---|---|---|
| the south facade frame with its parapet (top +8.00) · **V8.0, south wing v2** | 100 % | 100 % | 100 % |
| the same frame without a parapet (solid top +7.20) | 100 % | 100 % | 100 % |

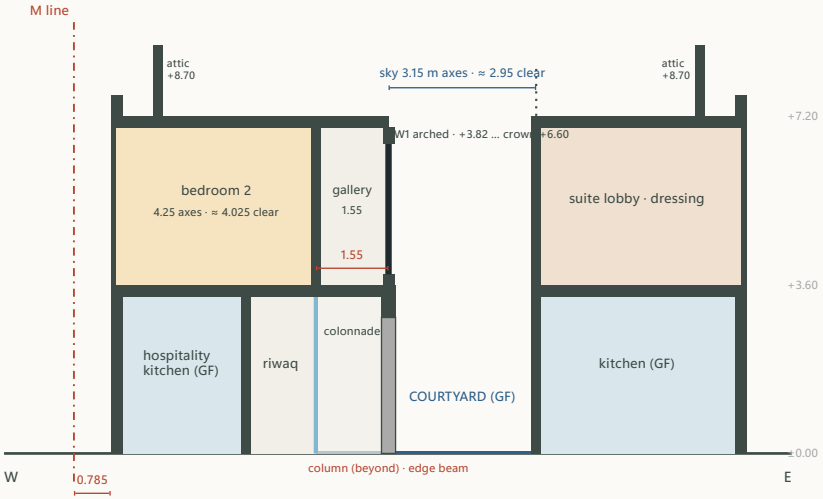
**How the percentages are defined.** “Sun on the windows”: the share of the window height (sill +5.00 to head +6.00) that receives direct sun at solar noon on 21 Dec (winter) or 21 Mar / 23 Sep (equinox) in this section; the ray from each point of the window must pass over the south facade’s parapet 4.75 m away (V8.0: the suite court runs to the facade). “Sky seen”: from a seated eye 1.2 m high in the double-height living (CSV y 16.1), the share of the windows’ vertical angle that shows sky rather than the facade. Pure geometry: no reflections, no sky model; a daylight simulation comes later.

- **The main light of the double-height living comes from the courtyard.** The 2 small south windows add light and let warm air out high on the south.
- **The 4.75 m in front of them is the suite court**, walled on four sides and open to the sky; the gym’s north wall is its south side, ending in a **railing, not a parapet**.
- **Privacy:** the windows open onto the suite court, so their sill stays above eye level from the court floor (or obscured glass); to check against the floor levels.
- **Summer:** the sun is almost overhead; light-coloured court walls and small hoods over the windows.

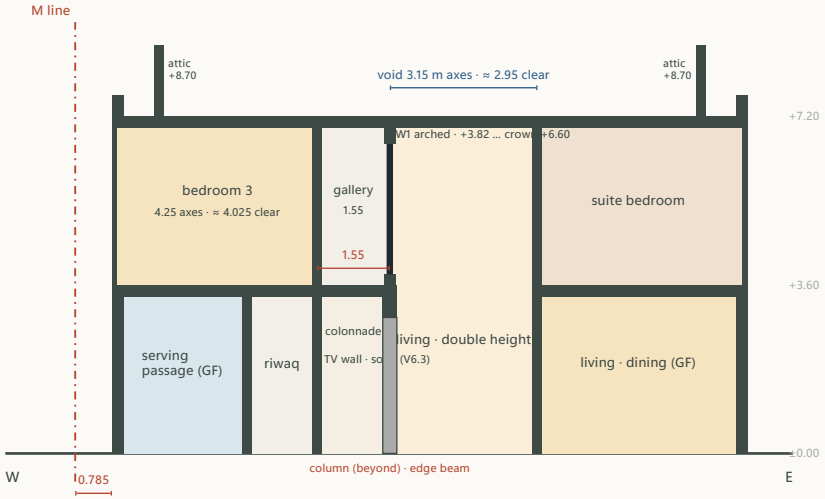
# First floor V8.1 · the west projection on three columns

sections looking north, CSV x · the column on x 5.8 is beyond the cut (columns at y 6.5, 13, 19.25)

### A–A · through the courtyard (y 12.0)




### B–B · through the double-height living (y 16.5)




| Item | Split projection (V5.1–V6.0, reverted) | V6.2 · 1.35 m inward on columns | Change |
|---|---|---|---|
| Courtyard sky opening, axes (≈ clear) | 4.025 (3.825) m | 3.15 (2.95) m | −0.875 |
| Double-height void width, axes | 4.025 m | 3.15 m | −0.875 |
| West room width, axes (≈ clear) | 4.25 (4.025) m | 4.25 (4.025) m | = |
| Gallery width, axes (≈ clear) | 1.35 (1.225) m | 1.55 (1.42) m | +0.2 |
| Columns in the courtyard / living | none · 0.675 m cantilever | **3** on x 5.8 + edge beam | +3 |
| FF west face → M line | 0.11 m (approval needed) | **0.785 m** (on the GF wall) | +0.675 m |
| Walls on cantilever beams | FF west wall, 14.95 m | none | – |
| FF outline (axes) | 318.86 m² | 309.68 m² | −9.176 m² |
| Courtyard sky + double-height void | 51.32 m² | 40.16 m² | −11.158 m² |
| FF floor (outline − voids) | 267.54 m² | 269.52 m² | +1.981 m² |
| · family bath | 3.74 m² (2.075 wide) | 9.42 m² (4.25 wide) | +5.685 m² |

V6.3 read from data/rooms_ff.csv; the split projection (V5.1–V6.0) from tag v6.0. Clear ≈ axes − ½ wall each side (250 outer, 200 inner, ≈ 50 mm for the steel window W1 on the void edge, V6.4); sky opening clear = axes − ½ of the east wall − the slab edge with its upstand.

- **Projection:** 1.55 m over the courtyard and the double-height void; the west rooms (4.25 m) and the gallery (1.55 m) take the band over the GF west band and riwaq.
- **Structure:** an edge beam on x 5.8 (spans 6.5 and 6.25 m) on 3 columns, each in an existing wall or glass line: the women's majlis south wall, the courtyard / living glass, the living's south wall. On the ground floor it reads as a colonnade along the courtyard. Walls stand on walls; nothing hangs on cantilever beams.
- **V6.1:** the split projection of V5.1–V6.0 (0.68 m in + 0.68 m out, the west wall on cantilever beams 0.11 m from the M line) is reverted: a wall on cantilever beams along ≈ 14.95 m costs more and is less direct than walls on walls, and the M-line projection needed municipal approval.
- **Cost of the revert:** courtyard sky 4.025 → 3.15 m; 3 columns on the courtyard edge.